Getting Started with NHANES in R: Data Structure, Data wrangling, and Visualizations

Use NHANES documentation, relative paths, XPT imports, SEQN joins, and beginner ggplot figures in one organized R workflow.

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Workshop agenda: NHANES data structure and R demo

5 min	**Workshop goals and NHANES overview**	What NHANES measures and why nutrition researchers use it
10 min	**Finding and organizing NHANES files**	Cycles, documentation, file families, data/raw, and data/processed
15 min	**R demo: import and join**	relative_path, read_xpt(), select(), left_join()
15 min	**Visualization**	ggplot checks and ggsave()
15 min	**Practice and wrap-up**	build one plot; What to check before real analysis

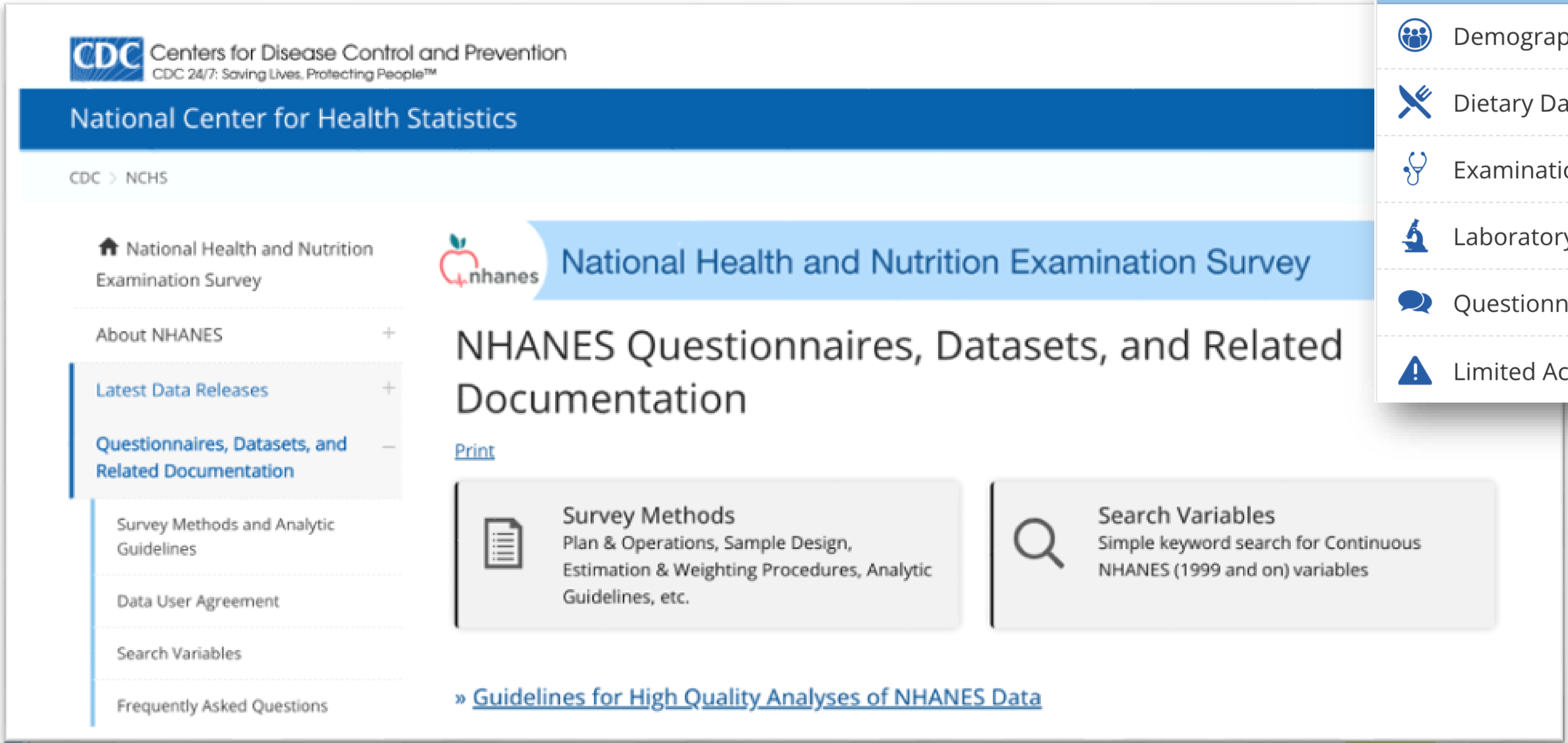
nhanes™

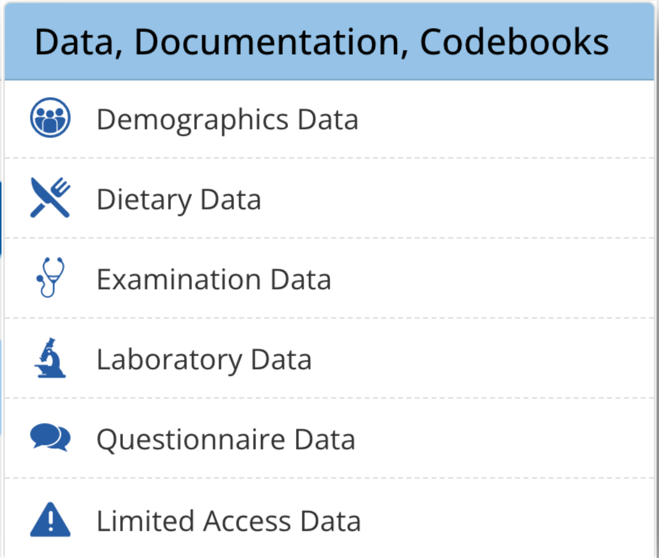
National Health and Nutrition Examination Survey

https://www.cdc.gov/nchs/nhanes/index.html

NHANES connects interviews, exams, labs, and diet

NHANES is a CDC/NCHS program that combines interviews, physical examinations, laboratory measures, and dietary information.





What is an NHANES cycle?

An NHANES cycle is a public-use data-collection period.

- Traditionally released in approximately two-year periods
- Each cycle contains a new cross-sectional sample
- A cycle does **not** follow the same participants over time
- File names identify which release the files belong to
- Match cycle markers when merging files

Reading NHANES File Names:

File stem + cycle marker

- **1999–2000:** DEMO — _A is implied but omitted
- **2001–2002:** DEMO_B
- **…**
- **2017–2018:** DEMO_J
- **Special prefix 2017 – March2020 pre-pandemic: P_DEMO**
- **August 2021–August 2023:** DEMO_L

NHANES Data Categories

Demographics
age, sex, race/ethnicity, income, survey design variables

Dietary
24-hour recalls, nutrient intake, supplement use

Examination
Body measures, blood pressure, dental and physical measures

Laboratory
Biomarkers from blood, urine, and other specimens

Questionnaires
health history, behaviors, medications, food security

Limited Access
Restricted geographic or sensitive variables

NHANES File Families and Units of Record

Family	Typical files	Unit of record / timing
Demographics	DEMO	Person; one record per participant per cycle
Dietary	DR1TOT/DR2TOT; DR1IFF/DR2IFF; supplement files	Person-day for totals; person-food for individual foods; Day 1 and Day 2 recalls; *Often requires aggregation or scoring, such as calculating a person-level HEI score.*
Examination	BMX, BPX	Person or examination visit; repeated readings may be separate variables
Laboratory	GLU, HDL, TRIGLY, TCHOL	Person/specimen; generally measured at the examination; some use subsamples
Questionnaire	DIQ, PAQ, DBQ	Person; reference period varies by question

Before merging, ask: What does one row represent, who was eligible, and which weight applies?

Why NHANES is useful for nutrition research

NHANES is one of the most widely used public health datasets for nutrition and epidemiology research.

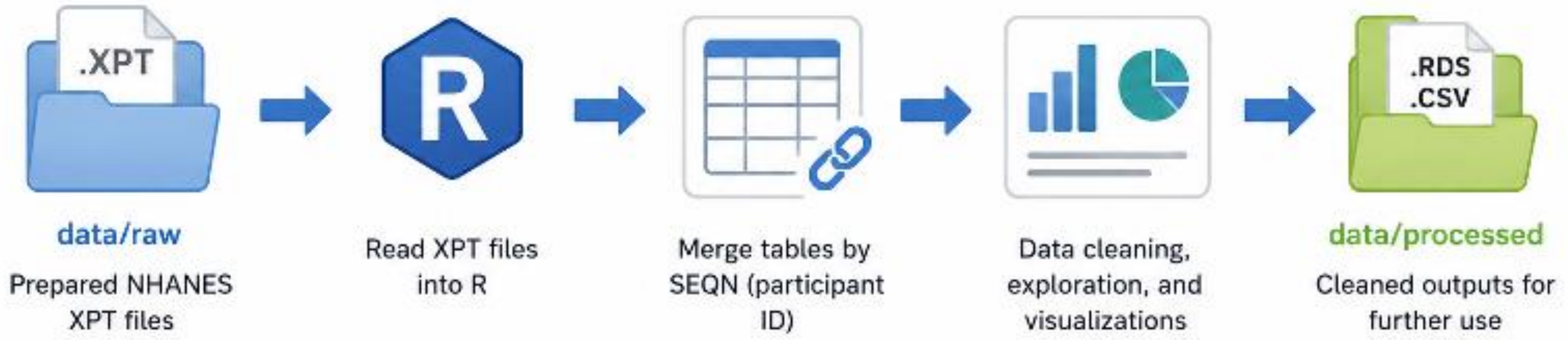
It support projects involving:

- Dietary intake
- Biomarkers and laboratory measures
- Anthropometrics (height, weight, BMI, etc.)
- Health conditions and risk factors

Always starts with:

✓ Survey cycle
✓ Codebook
✓ Eligibility criteria
✓ Missing-value codes
✓ Appropriate survey weights

Today's workflow



In today's demo, we use prepared XPT files in **data/raw** and write cleaned outputs to **data/processed**.

The live R code demonstrates the workflow rather than a full survey-weighted analysis.

Resources:

- About NHANES: https://www.cdc.gov/nchs/nhanes/about/index.html
- NHANES datasets and documentation: https://wwwn.cdc.gov/nchs/nhanes/default.aspx
- Continuous NHANES: https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/
- NHANES tutorials: https://wwwn.cdc.gov/nchs/nhanes/tutorials/default.aspx

Treat documentation as part of the dataset

A clean project makes it easy to rerun analysis and explain every recode.

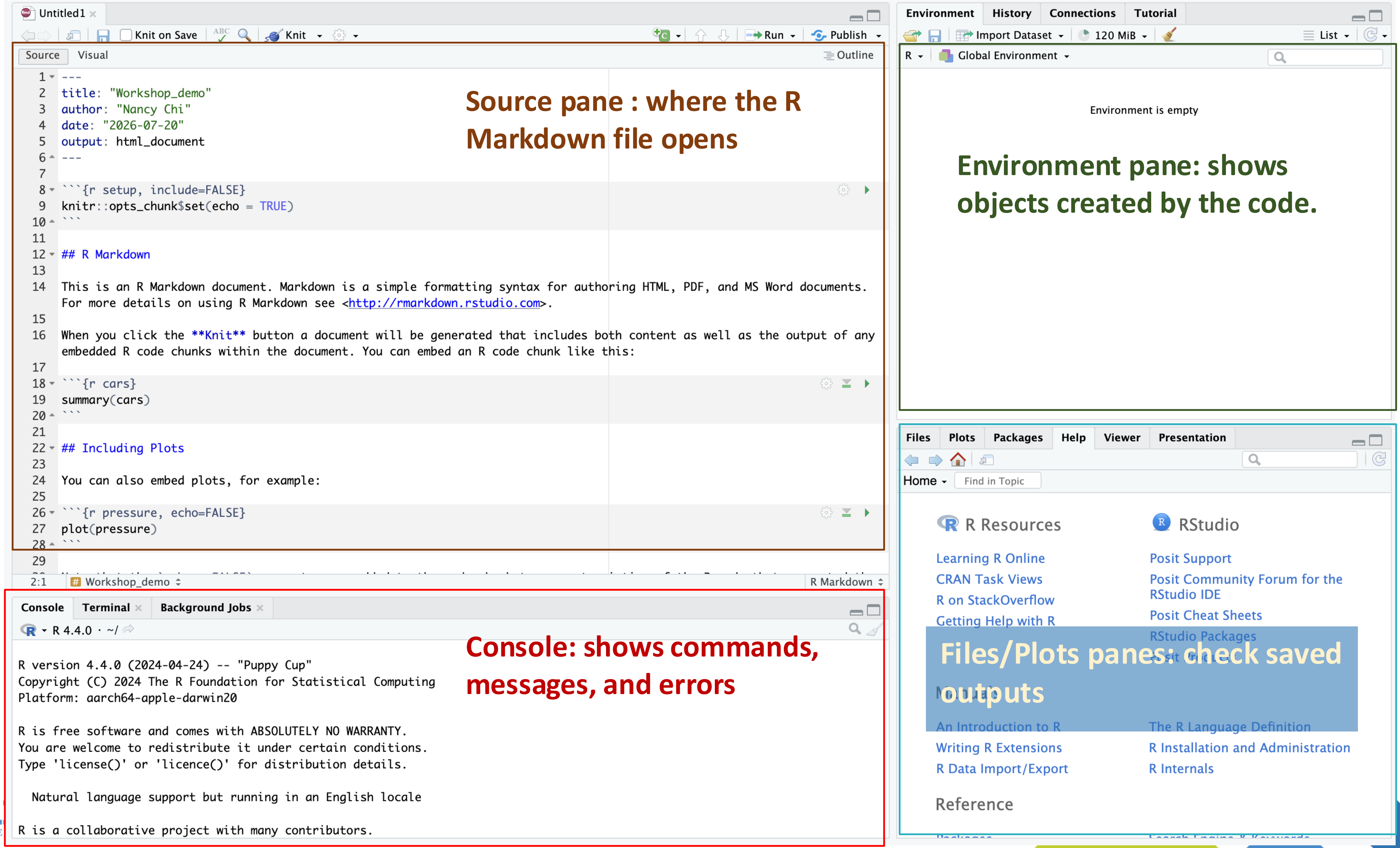
```
NHANES_project/
├── data/
│   ├── raw/              Original, untouched NHANES XPT files
│   └── processed/        Cleaned, merged, or analysis-ready datasets
├── code/                 R Markdown code file
├── docs/
│   └── documentation/    Saved CDC codebooks and documentation
└── outputs/
    └── figures/          Saved plots, tables, and other outputs
```

Data File Name	Doc File	Data File	Date Published
Demographic Variables and Sample Weights	DEMO_J Doc	DEMO_J Data [XPT - 3.3 MB]	February 2020

Before coding

1. Confirm the cycle
2. Save the Data file and Doc File
3. Read eligibility and missing-value notes
4. Identify the join key: SEQN
5. Identify the correct survey weight

RStudio set up for the live R Markdown demo



Most participant-level joins use SEQN as the respondent identifier

DEMO_J

SEQN, age, sex, weights, design

left_join

BMX_J

SEQN, height, weight, BMI

left_join

DR1TOT_J

SEQN, day 1 energy, protein

- Codebook
  - SEQN - Respondent sequence number
  - SDDSRVYR - Data release cycle
  - RIDSTATR - Interview/Examination status
  - RIAGENDR - Gender
  - RIDAGEYR - Age in years at screening
  - RIDAGEMN - Age in months at screening - 0 to 24 mos

Rule of thumb: inspect row counts and duplicate SEQN values before every join. In the demo, SEQN is the shared participant identifier across DEMO_J, BMX_J, and DR1TOT_J.

Joining tables and combining cycles are different operations

Join within a cycle (demo)

Use when columns live in different files for the same participants.

R verb: left_join(..., by = "SEQN")

Append across cycles

Use when rows come from different survey periods after harmonizing variables.

R verb: bind_rows()

For population estimates, pooled-cycle weights need explicit attention.

dplyr verbs you can map to Stata/Python habits

Task	R / dplyr	Stata mental model	Python / pandas
Keep variables	select(SEQN, RIDAGEYR, BMXBMI)	keep SEQN RIDAGEYR BMXBMI	df[["SEQN", "RIDAGEYR", "BMXBMI"]]
Filter rows	filter(RIDAGEYR >= 20)	keep if RIDAGEYR >= 20	df[df["RIDAGEYR"] >= 20]
Create variable	Mutate(adult = RIDAGEYR >= 20)	gen adult = RIDAGEYR >= 20	df.assign(adult = df["RIDAGEYR"] >= 20)
Merge files	left_join(bmx, by = "SEQN")	merge 1:1 SEQN using bmx	df.merge(bmx, on = "SEQN", how = "left")
Append cycles	bind_rows(cycle1, cycle2)	append using cycle2	pd.concat([cycle1, cycle2], ignore_index = TRUE)

A first ggplot should answer a data-checking question

Demo plot 1: BMI by age, colored by sex.
Demo plot 2: Day 1 energy intake by BMI group.
Save both figures with ggsave().

Ask before polishing

- Is the variable plausible?
- Are missing values visible?
- Are groups coded correctly?
- Does the plot match the research question?

Survey design matters before reporting estimates

For descriptive population estimates, NHANES analyses usually require the correct sample weight, strata, and PSU variables. Today we name these design elements; you may explore survey-weighted estimation on your own.

Weight

Which participants and measurement component does this represent?

Strata

Which design strata define variance estimation?

PSU

Which primary sampling units define clustering?

In-class activity: modify the BMI-by-age plot

01 Open the In-Class Practice section in the Rmd

02 Change color to BMI group

03 Add facet_wrap(~ sex)

04 Save practice_bmi_by_age.png

05 Write one pattern and one limitation

A credible NHANES analysis starts before modeling

Question
What population, cycle, and measurement domain?

Documentation
Which codebook, eligibility notes, and missing codes?

Assembly
Which joins by SEQN and which appends by cycle?

Design
Which weight, strata, and PSU for the estimate?

In NHANES, the quality of the final estimate depends as much on how we define, document, and assemble the data as it does on the model we choose.

Thank you

Demo handout: demo/nhanes_workshop_demo.html
R Markdown source: demo/nhanes_workshop_demo.Rmd
CDC data index: wwwn.cdc.gov/nchs/nhanes